Hybrid Machine-Learning / Molecular-Mechanics Energy Functions for Condensed-Phase Simulations: Training and Molecular Dynamics

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Abstract

We describe a hybrid machine-learning / molecular-mechanics (ML/MM) energy function for condensed-phase simulations in which intramolecular and short-range intermolecular interactions are evaluated with a neural-network potential, while intermolecular Lennard-Jones and Coulomb interactions are supplied by a CHARMM/CGenFF-compatible classical term that is switched on as a function of monomer center-of-mass separation. The same hybrid total energy is assembled during supervised training and during molecular dynamics (MD), so that the learned model is trained on the Hamiltonian later used for dynamics. Mechanical and electrostatic embeddings for the classical Coulomb term are controlled by a charge-mode switch. Periodic long-range electrostatics are provided by optional Ewald / particle-mesh Ewald solvers. We outline validation targets—bonded MM parity, NVE energy conservation, rigid dimer scans, liquid densities, and free-energy profiles from umbrella sampling and CHARMM adaptive umbrella dynamics (ADUMB)—and provide regenerable workflow hooks; nu-

merical panels that are not yet frozen appear as explicit placeholders. Implementation details and CLI examples are collected in the Supporting Information.

Contents

1 Introduction

Machine-learned interatomic potentials have transformed the accuracy–cost trade-off for molecular simulation, from early high-dimensional neural network representations[1,2] and Gaussian approximation potentials[3,4] to message-passing architectures such as SchNet,[5] PhysNet,[6] SpookyNet,[7] and related force fields trained to near-quantum fidelity.[8–11] Nonetheless, treating large condensed-phase systems entirely with neural networks remains expensive when long-range electrostatics and many solvent molecules must be retained.

Classical molecular-mechanics (MM) force fields such as CHARMM/CGenFF[12–14] remain efficient for intermolecular interactions at longer range, and particle-mesh Ewald (PME) methods provide scalable periodic Coulomb sums.[15–17] Hybrid schemes that combine a local high-accuracy model with an efficient environment have a long history in QM/MM chemistry[18–20] and motivate analogous ML/MM constructions: ML for intramolecular and short-range many-body interactions, MM for the intermolecular tail and optional long-range Coulomb solvers.[6,21]

A practical hybrid scheme must expose a single total energy both when fitting parameters and when integrating equations of motion. The MMML package[21] implements that requirement through a shared hybrid forward pass used by training and MD entry points. The remainder of this article defines the hybrid energy (Section 2), summarizes training and condensed-phase protocols (Sections 3–4), describes charge embeddings (Section 5), and presents the validation and application targets that the accompanying Snakemake workflows are designed to regenerate (Section 6). Software command inventories and YAML templates are deferred to the Supporting Information.

2 Hybrid ML/MM Energy Function

2.1 Monomer Partition and Center-of-Mass Switching

Interactions are partitioned by monomer (typically one solvent residue). Let r_{IJ} denote the center-of-mass (COM) distance between monomers I and J. The hybrid potential takes the form

$$E = \sum_I E_I^{\mathrm{ML}} + \sum_{I<J} \left[s_{\mathrm{ML}}(r_{IJ})\, E_{IJ}^{\mathrm{ML}} + s_{\mathrm{MM}}(r_{IJ})\, E_{IJ}^{\mathrm{MM}} \right] + E^{\mathrm{LR}} + E^{\mathrm{wall/restr}}. \tag{1}$$

where E_I^{ML} is the intramolecular ML energy of monomer I, E_{IJ}^{ML} is the ML two-body (dimer) correction, and E_{IJ}^{MM} collects switched intermolecular Lennard-Jones and Coulomb contributions evaluated with CGenFF-compatible parameters. Optional long-range Coulomb beyond the real-space switch enters as E^{LR} (Ewald,[15] PME,[16,17] or related solvers). A short-range steric wall and restraints may contribute $E^{\mathrm{wall/restr}}$ (including optional Ziegler–Biersack–Littmark nuclear repulsion inside the neural model[22]).

For an isolated dimer AB, Equation 1 is equivalent to the training assembly

$$E = (1 - s)\,(E_A + E_B) + s\,E_{AB} + E_{\mathrm{MM}}, \tag{2}$$

with $s = s_{\mathrm{ML}}(r_{\mathrm{COM}})$, $\Delta E_{\mathrm{ML}} = E_{AB} - E_A - E_B$, and $E = E_A + E_B + s\,\Delta E_{\mathrm{ML}} + E_{\mathrm{MM}}$. The taper multiplies only the dimer interaction; monomer intramolecular energies remain fully on at all separations. Forces include the product-rule term arising from $\partial s/\partial \mathbf{R}$.

2.2 Default Handoff Window

With complementary handoff, $s_{\mathrm{ML}} + s_{\mathrm{MM}} = 1$ inside the ML↔MM transition. Production defaults used throughout training and MD YAML are listed in Table 1 and illustrated in

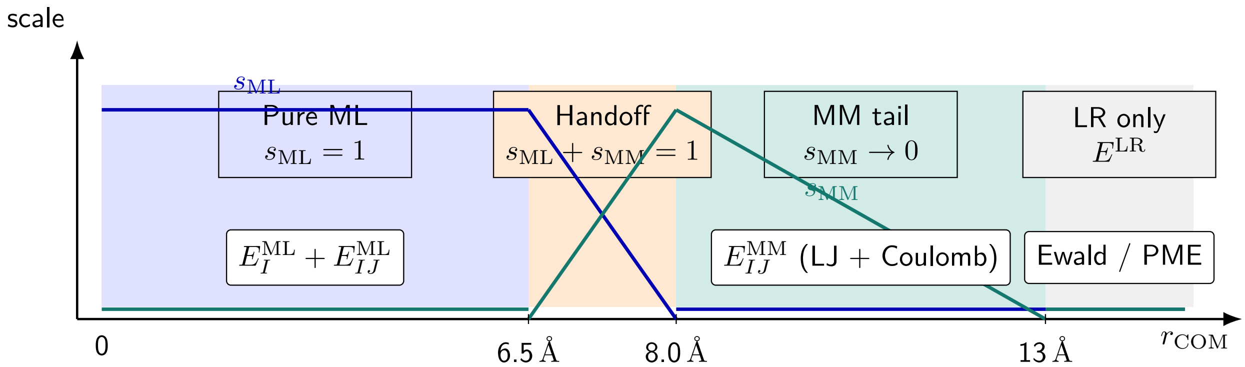

Figure 1: Schematic COM handoff for the hybrid energy in Equation 1. Below ≈ 6.5 Å the intermolecular ML dimer term is fully on; between 6.5 Å and 8.0 Å ML and MM scales are complementary; the switched MM pair term decays by 13 Å. Beyond the MM cutoff, only optional long-range Coulomb (E^{LR}) remains.

Figure 1.

Table 1: Default COM handoff parameters shared by hybrid training and MD.

Parameter	Symbol / role	Default
`mm_switch_on`	end of ML→MM handoff	8.0 Å
`ml_switch_width`	ML taper width	1.5 Å
`mm_switch_width`	MM outer tail	5.0 Å

2.3 Ownership of Terms

Table 2 summarizes which component owns each contribution.

3 Hybrid Training

Hybrid models are trained so that the supervised target is the hybrid total of Equation 2, not the bare neural energy. Datasets are covalent dimers with QM labels (R, Z, E, $\mathbf{F}$, optional multipoles)[24–27] augmented by CGenFF type indices and charges. The objective is a weighted sum of squared residuals on hybrid energies and forces (and optional dipole /

Table 2: Term ownership in the hybrid Hamiltonian.

Term	Owner	Notes
E_I^{ML}	PhysNet / Spooky / SO3LR[6,7,23]	Always on
E_{IJ}^{ML}	PhysNet / Spooky / SO3LR	COM-gated by s_{ML}
LJ + Coulomb pairs	CGenFF MM	Scaled by s_{MM}; q from charge mode
Optional LJ scales	Trainable s^σ, s^ε	Sidecar scales file
E^{LR}	Ewald / PME / ...	Long-range solver flag
Bonded CHARMM	Topology (Mode B)	Or jax-mm-spoof bonded clone

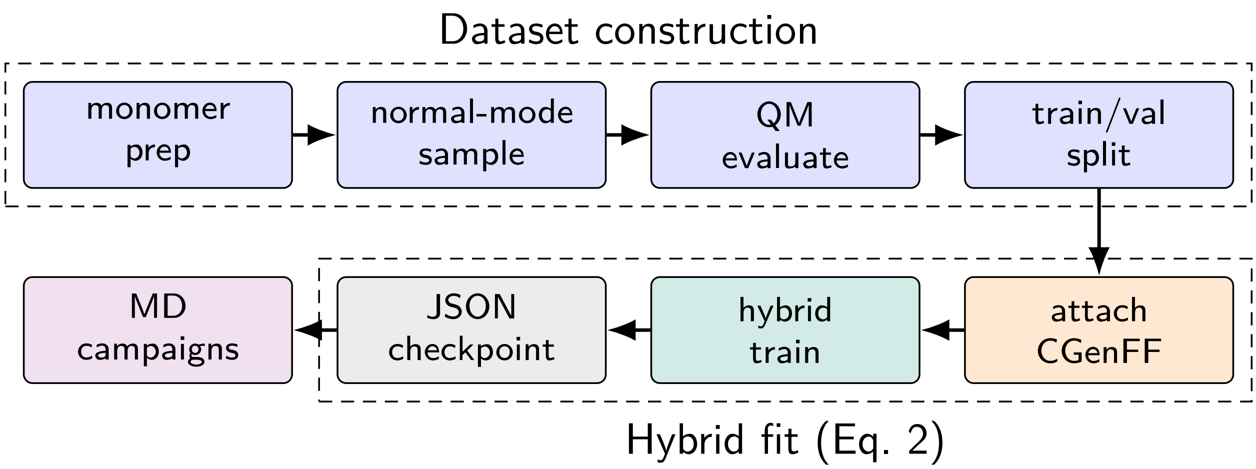

Figure 2: Hybrid training pipeline. Quantum labels on dimer geometries are augmented with CGenFF arrays; hybrid training fits Equation 2. The resulting portable JSON checkpoint is consumed unchanged by MD campaigns.

charge heads),

$$\mathcal{L} = w_E \|E - E^{\mathrm{ref}}\|^2 + w_F \|\mathbf{F} - \mathbf{F}^{\mathrm{ref}}\|^2 + w_D \|\boldsymbol{\mu} - \boldsymbol{\mu}^{\mathrm{ref}}\|^2 + w_q \|q - q^{\mathrm{ref}}\|^2. \tag{3}$$

Representative weights in the bundled fixed-charge template are $w_E = 1$, $w_F = 52.91$, $w_D = 27.21$, and $w_q = 14.39$. The PhysNet radial cutoff must be at least the MM handoff onset (Table 1). Portable JSON checkpoints (and optional LJ-scale sidecars) are written for MD deployment. The dataset → fit → MD pipeline is summarized in Figure 2; CLI flags and invocation examples are collected in Section S1.

4 Condensed-Phase Molecular Dynamics

4.1 Two Drivers, One Hamiltonian

Dynamics are organized as campaign YAML files that chain box preparation, minimization, heating, equilibration, and production under a shared set of handoff and long-range defaults (Supporting Information, Section S2).[28,29] Supported backends are PyCHARMM (CHARMM integrator with an MLpot USER callback),[12] jax-md (pure-JAX hybrid energy with a jax-md integrator),[30] and ASE for reference smokes. Thermostats and barostats follow standard Nosé–Hoover / MTTK practice.[31,32]

- Mode A (jax-md). CHARMM is used only to build the topology; every energy term is evaluated in a jitted hybrid energy function.
- Mode B (PyCHARMM). CHARMM advances the trajectory; the hybrid ML contribution enters each step through the MLpot callback while classical bonded / nonbonded bookkeeping follows the PSF.

Both drivers evaluate Equation 1 with the same handoff parameters and charge mode.

4.2 Box Preparation and Dynamics Stages

Dense liquids follow a two-phase protocol (Figure 3):

1. Phase A (MM certify). Packmol placement,[33] optional Monte Carlo density moves, and classical minimization / short dynamics until inter-monomer contact criteria pass.
2. Phase B (hybrid MD). Register the ML potential (or a jax-mm-spoof bonded clone for infrastructure tests), minimize under the hybrid energy, then heat $\rightarrow$ equilibrate $\rightarrow$ produce.

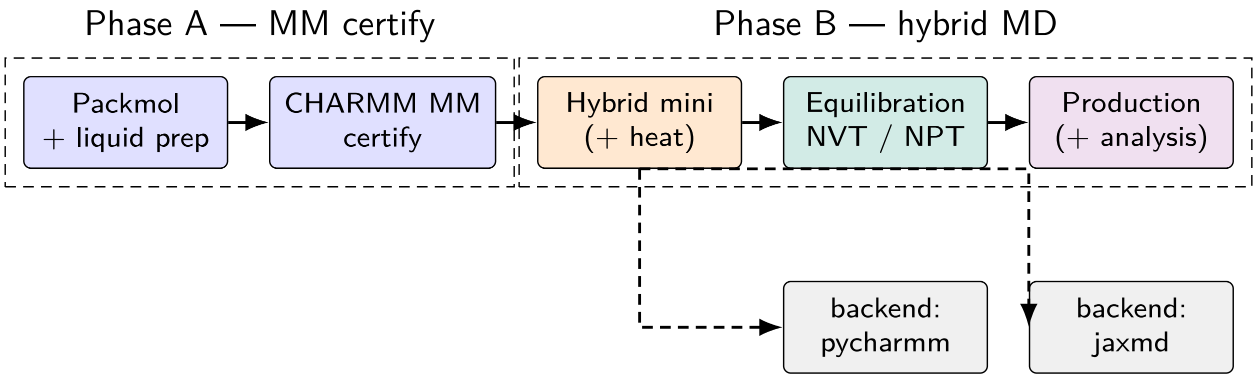

Figure 3: Condensed-phase campaign pipeline. Phase A builds and certifies a classical liquid box; Phase B registers the hybrid potential and runs mini/heat/equilibration/production on either the PyCHARMM or jax-md backend.

4.3 Free-Energy and Scanning Protocols

Beyond equilibrium liquid MD, the same hybrid energy supports:

- Rigid dimer / pair scans along a COM or atom–atom distance to validate the ML↔MM handoff against QM or DES reference curves;
- Umbrella sampling with harmonic restraints $W_k = \frac{1}{2}k_k(\xi - \xi_{0,k})^2$ and MBAR analysis[34,35] (gas-phase packed ML windows or hybrid mechanical embedding in explicit solvent);
- CHARMM ADUMB adaptive umbrella / WHAM paths[36,37] driven through PyCHARMM reaction-coordinate lingo for reactive complexes (e.g. Menshutkin-type $NH_3 + CH_3Cl$ studies[38,39]).

CLI entry points and example YAML are given in Sections S2 and S3.

5 Mechanical and Electrostatic Embeddings

The classical Coulomb term inside E_{MM} obtains its per-atom charges from a charge-mode switch. This choice is orthogonal to whether the neural model predicts charges for use inside E^{ML}.

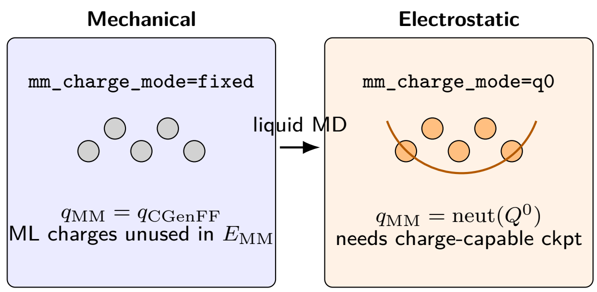

Figure 4: Liquid-capable embeddings for hybrid MM Coulomb. Mechanical embedding keeps fixed CGenFF charges in E_{MM}. Electrostatic embedding (Q^0) rebuilds neutralized monomer ML charges each evaluation.

Mechanical embedding (**`fixed`**). $q_{\mathrm{MM}} = q_{\mathrm{CGenFF}}$. Default for charge-less PhysNet checkpoints and for LJ-fitting stages.

Electrostatic embedding (**`q0`** / Q^0). $q_{\mathrm{MM}} = \mathrm{neutralize}(Q^0)$, where Q^0 denotes unperturbed monomer charge-head evaluations. Requires a charge-capable checkpoint.

Dimer-only modes (**`latent`**/Q^1). Partner-perturbed charges from the AB forward; undefined for $N > 2$ monomers and therefore not used in liquid production matrices.

Figure 4 contrasts the two liquid-capable embeddings.

6 Results and Illustrative Applications

Unless noted, panels marked (PLACEHOLDER) will be replaced by figures exported from versioned Snakemake workflows under `workflows/` (artifact paths and provenance checklist in Section S4). Numbers already available in-repository are filled in; empty cells await frozen production runs.

6.1 Bonded MM Parity (jax-mm-spoof)

The hybrid jax-md path can replace the neural model by a JAX CGenFF bonded clone (`jax_mm_spoof`) so that drivers can be certified without a trained checkpoint. Bonded

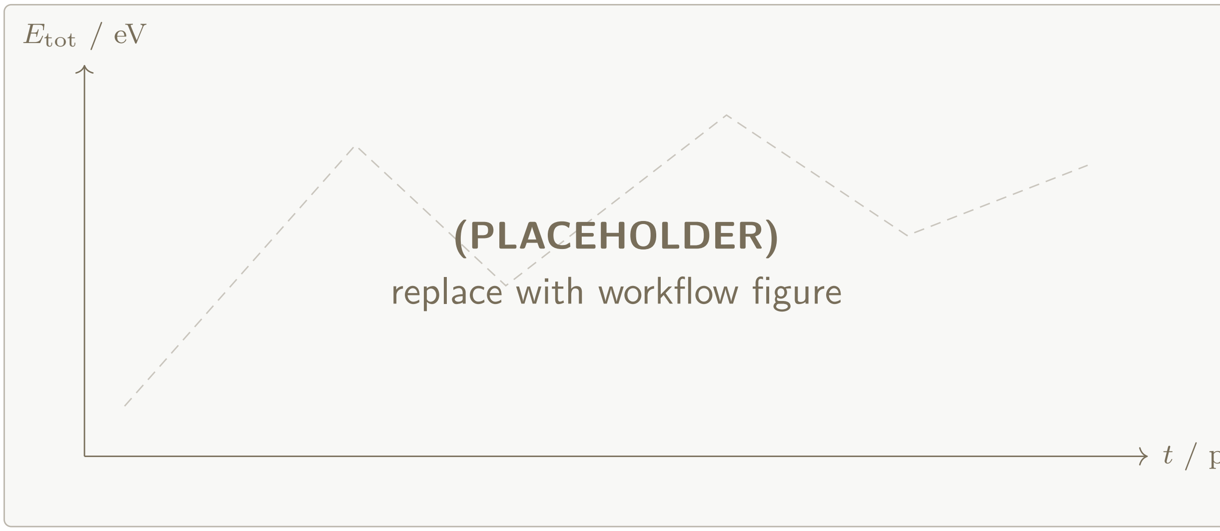

Figure 5: (PLACEHOLDER) NVE total-energy traces for hybrid DCM under representative COM cutoff presets (workflow `dcm3_nve_cutoff_sweep`). Report mean drift ($\mathrm{eV\,ps^{-1}}$ or $\mathrm{kcal\,mol^{-1}\,ps^{-1}}$) and peak-to-peak noise in the accompanying metrics table.

energy components for dichloromethane (DCM) and acetone (ACO) monomers match native PyCHARMM to machine precision (Table 3; workflow `jaxmd_cgenff_spoof_smoke`).

Table 3: Bonded energy parity: jax-mm-spoof versus native CHARMM ($\mathrm{kcal\,mol^{-1}}$).

Case	E_{jax}	E_{CHARMM}	ΔE
DCM fixture	311.3070436736117	311.30704367361164	$+5.7 \times 10^{-14}$
DCM smoke-min	1.6671786898619212	1.667178689861924	-2.9×10^{-15}
ACO fixture	7.779347176527917	7.779347176527915	$+1.8 \times 10^{-15}$
ACO smoke-min	1.9927947535063575	1.9927947535063568	$+6.7 \times 10^{-16}$

6.2 Energy Conservation (NVE)

Integrator and handoff robustness are assessed in the microcanonical ensemble by monitoring total-energy drift and high-frequency noise as a function of COM cutoff preset and system size (`dcm3_nve_cutoff_sweep`, `dcm_nve_scaling`). Figure 5 will show representative $E(t)$ traces for hybrid DCM clusters / liquid boxes under jax-md and PyCHARMM.

Table 4: (PLACEHOLDER) NVE conservation metrics for hybrid DCM (fill from sweep `metrics.json`).

Preset	Backend	N_{mol}	$\langle \Delta E \rangle$	RMS noise	Status
paper-default	jaxmd	—	—	—	pending
paper-default	pycharmm	—	—	—	pending
tight / loose	jaxmd	—	—	—	pending

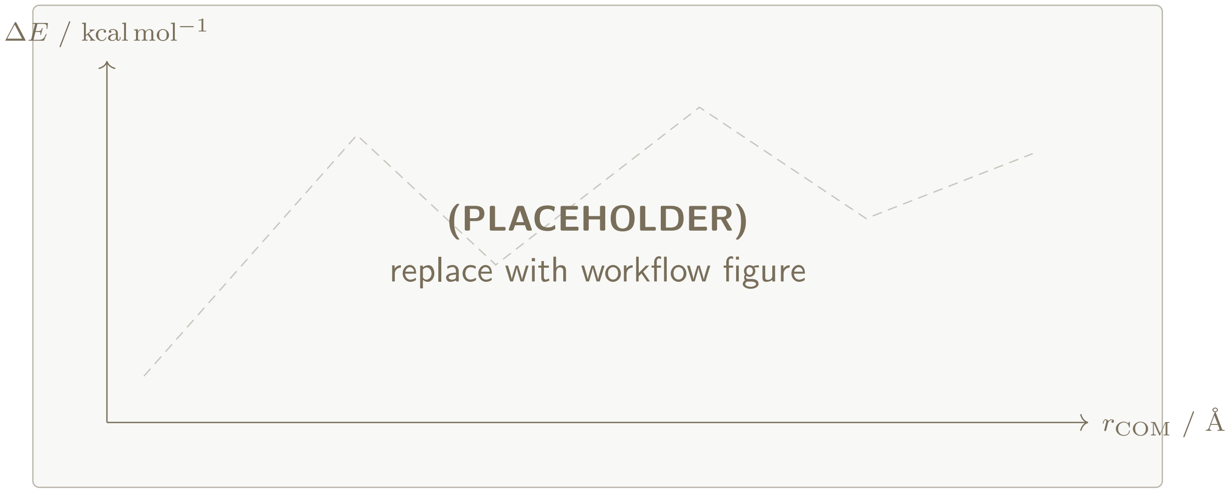

Figure 6: (PLACEHOLDER) Rigid dimer interaction scan: reference versus ML, MM, and hybrid (Equation 2) along r_{COM}. Vertical guides mark the default handoff window of Table 1.

6.3 Rigid Dimer and Pair Scans

Rigid scans along intermolecular separation probe whether the hybrid handoff reproduces reference interaction curves outside the fitting set. Primary sources are DES-style dimer panels (`des_dimer_pair_scans`) and solvent-specific COM scans. Figure 6 contrasts ML-only, MM-only, and hybrid $E(r)$ against a QM / DES reference.

6.4 Liquid Densities

Dichloromethane and acetone liquids are targeted under NPT/NVT hybrid dynamics (`pbc_liquid_density` methane at fixed liquid-like density with `lr_solver: ewald` is covered by `pbc_methane_ewald`. Table 5 and Figure 7 summarize the planned reporting matrix. Experimental liquid densities are listed as orientation only until production $\langle \rho \rangle$ values are frozen.

Table 5: (PLACEHOLDER) Liquid densities under hybrid MD ($T = 300\,\mathrm{K}$ unless noted).

Solvent	N	Ens.	Backend	Emb.	$\langle\rho\rangle$ ($\mathrm{g\,cm^{-3}}$)	Expt. ($\mathrm{g\,cm^{-3}}$)
DCM	—	NPT	jaxmd	mech.	—	~1.33
DCM	—	NPT	pycharmm	mech.	—	~1.33
ACO	—	NPT	jaxmd	m./es.	—	~0.78
CH_4	—	NVT	both	m./es.	(input)	—

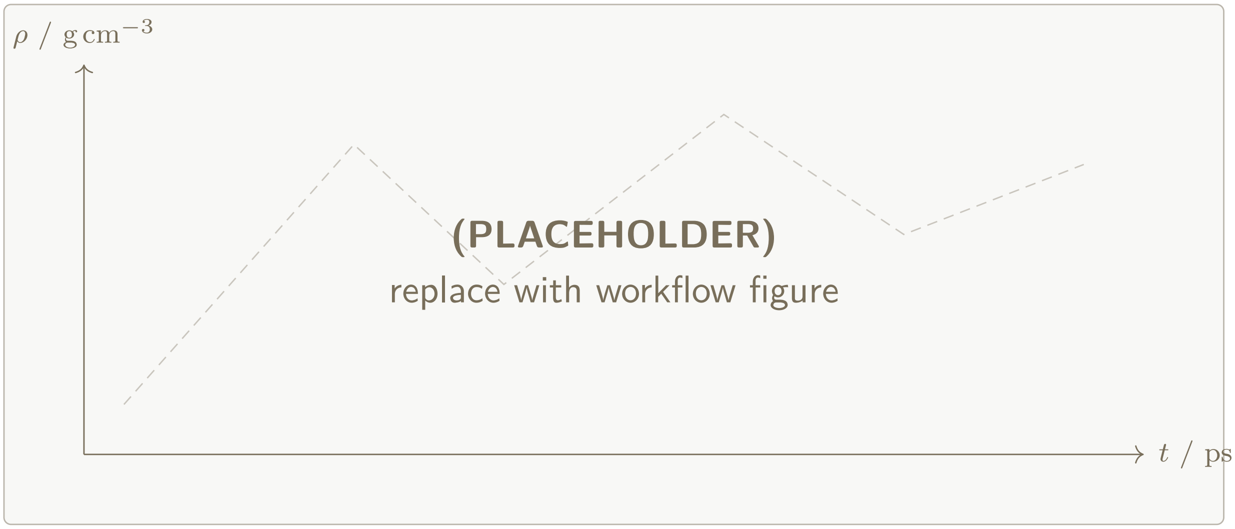

Figure 7: (PLACEHOLDER) Density time series $\rho(t)$ for hybrid DCM and ACO production windows (`pbc_liquid_density_dyn`). Horizontal lines: experimental liquid densities.

6.5 Umbrella Sampling and MBAR Free Energies

Gas-phase and explicit-solvent free-energy profiles are obtained from windowed umbrella sampling[34] with MBAR reweighting[35,40,41] (`mmml umbrella-sample` / `umbrella-mbar`). The Menshutkin-type $NH_3 + CH_3Cl$ reaction coordinate $\xi = r(\mathrm{C{-}Cl}) - r(\mathrm{C{-}N})$ is the primary reactive example.[38,39] Figure 8 will show window histograms and the MBAR PMF; Table 6 records window settings.

Table 6: (PLACEHOLDER) Umbrella window summary for the reactive PMF.

Engine	N_{win}	k	ξ range	T / K	Status
packed_ml	—	—	—	300	pending
hybrid_jaxmd	—	—	—	300	pending

Figure 8: (PLACEHOLDER) Umbrella-sampling PMF along ξ for the Menshutkin-type complex (gas-phase packed ML and/or hybrid mechanical embedding in TIP3P water[42]). Inset (future): overlapping window histograms.

6.6 CHARMM ADUMB Paths

Adaptive umbrella sampling in CHARMM (ADUMB / reaction-coordinate WHAM)[12,36,37] provides an independent free-energy route for hybrid PyCHARMM+MLpot systems. ADUMB is issued through pre-dynamics CHARMM lingo; outputs include bias histograms and WHAM profiles (`ADUMB-WUNI.DAT`, `UMBCOR`). Figure 9 will compare ADUMB and MBAR PMFs on a shared CV where both protocols apply.

6.7 Simulation Matrix

Table 7 lists the intended production cells for the first frozen manuscript bundle. Smoke-level engineering runs already exercise mechanical and electrostatic embeddings for methane/Ewald (DES and So3LR checkpoints) but are not promoted as property claims.

7 Conclusions

The hybrid ML/MM formulation in Equation 1 and the training assembly in Equation 2 provide a closed loop from dimer QM data to condensed-phase dynamics and free-energy

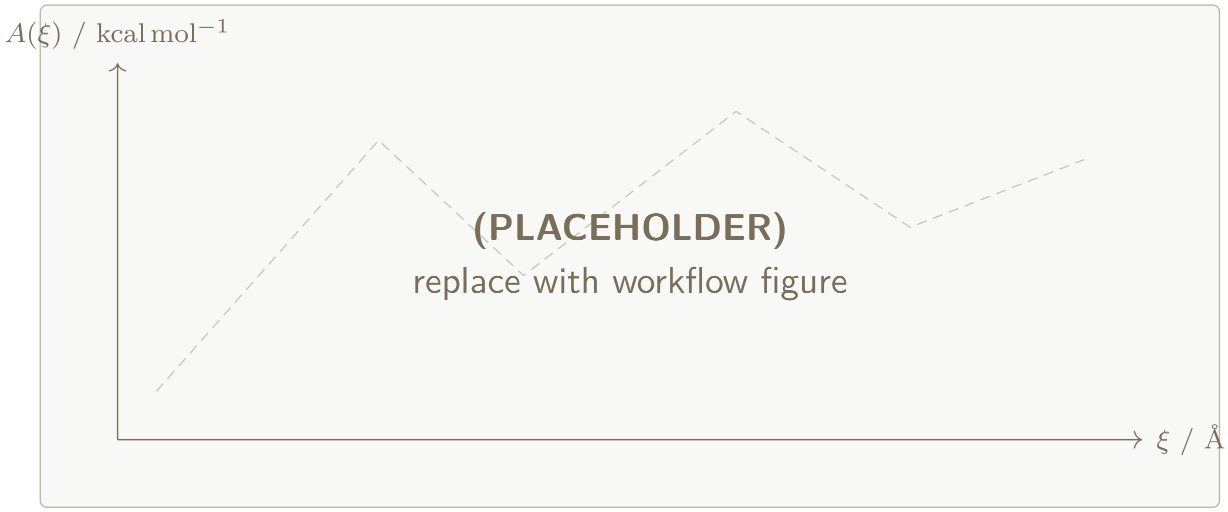

Figure 9: (PLACEHOLDER) CHARMM ADUMB free-energy profile for the hybrid reactive complex, optionally overlaid with the MBAR PMF of Figure 8. Workflow / example tree: `examples/m/` and `workflows/nh3_ch3cl_reaction_path/`.

Table 7: (PLACEHOLDER) Production simulation matrix for regenerable Results panels.

System	Protocol	Backend	Embedding	Workflow
DCM/ACO bonded	spoof parity	jaxmd	—	`jaxmd_cgenff_spoof_smoke`
DCM cluster	NVE cutoffs	both	mech.	`dcm3_nve_cutoff_sweep`
DCM/ACO liquid	NPT density	both	mech./es.	`pbc_liquid_density_dyn`
CH_4 liquid	NVT+Ewald	both	mech./es.	`pbc_methane_ewald`
NH_3/CH_3Cl	umbrella+MBAR	jaxmd	mech.	umbrella examples
NH_3/CH_3Cl	ADUMB	pycharmm	mech.	ADUMB examples

protocols. Mechanical and electrostatic embeddings select how classical Coulomb charges are obtained without changing ML/MM ownership of bonded and short-range many-body terms. Shared handoff defaults and portable JSON checkpoints keep training and MD numerically aligned across the jax-md and PyCHARMM drivers. The Results panels above—energy conservation, rigid scans, liquid densities, umbrella/MBAR, and ADUMB—are structured so that each figure or table cites a regenerable workflow; placeholders will be replaced as production artifacts are frozen.

Acknowledgement

We thank members of the Meuwly group for discussions on hybrid ML/MM workflows. Computational resources at the University of Basel are gratefully acknowledged.

Supporting Information Available

Supporting Information is appended at the end of this file (CLI inventories, YAML templates, free-energy command examples, and workflow provenance notes). Extended CLI documentation also lives in the MMML documentation tree.

Supporting Information

Hybrid Machine-Learning / Molecular-Mechanics Energy Functions for Condensed-Phase Simulations

Eric Boittier and Markus Meuwly

S1 Training CLI and Flags

Hybrid models are trained with

```
mmml physnet-train --hybrid-mm [--config train.yaml]
```

Essential flags (YAML or CLI):

- `--hybrid-mm` — enable hybrid total of Equation 2
- `--mm-charge-mode` {`fixed`, `q0`, `latent`, ...}
- `--charges` / `--no-charges` — neural charge head
- `--ml-switch-width`, `--mm-switch-on`, `--mm-switch-width`
- `--lr-solver` {`mic`, `ewald`, `nvalchemiops_pme`, ...}
- `--learn-mm-lj-scales` — optional trainable LJ scales

CGenFF fields in the NPZ (`cgenff_type_idx`, `mol_id`, `cgenff_charge`, master σ/ε) are prepared after electronic-structure evaluation of normal-mode samples, e.g.

```
mmml prepare-mm-dataset ...
mmml physnet-train --hybrid-mm --config train.yaml
```

S2 MD Campaign CLI and YAML

Dynamics are launched with

```
mmml md-system --config campaign.yaml --run-all [--resume]
```

A campaign file supplies global `defaults` (composition, box, checkpoint, cutoffs, `lr_solver`, `mm_charge_mode`, backend options) and an ordered list of `runs/jobs`. Dependencies are resolved automatically; each leg writes a handoff state consumed by the next stage.

S2.1 Minimal campaign sketch

```yaml
defaults:
  composition: "DCM:64"
  backend: jaxmd
  setup: pbc_npt
  checkpoint: path/to/hybrid.json
  mm_switch_on: 8.0
  ml_switch_width: 1.5
  mm_switch_width: 5.0
  lr_solver: ewald
  mm_charge_mode: fixed
runs:
  - { id: mini, stage: minimize }
  - { id: heat, stage: heat, depends: [mini] }
  - { id: equi, stage: equilibrate, depends: [heat] }
  - { id: prod, stage: produce, depends: [equi] }
```

S2.2 Paper workflow entry points

- `workflows/pbc_liquid_density_dyn/` — DCM and ACO liquids

- `workflows/pbc_methane_ewald/` — methane + Ewald, embedding sweep
- `workflows/dcm3_nve_cutoff_sweep/` — NVE / cutoff ranking
- `workflows/jaxmd_cgenff_spoof_smoke/` — bonded parity
- `workflows/des_dimer_pair_scans/` — rigid DES panels

Table S1: Software stack for hybrid training and MD.

Component	Role
MMML[21]	CLI, hybrid calculator, workflows
JAX / jax-md[30]	Differentiable hybrid energy; jaxmd backend
PyCHARMM / CHARMM[12]	Topology, classical MM, pycharmm backend
PhysNet / SpookyNet / SO3LR[6,7,23]	Neural short-range models
Snakemake (+ Slurm)	Parameter sweeps and cluster submission

S3 Free-Energy Protocols (Umbrella and ADUMB)

S3.1 Batched / hybrid umbrella sampling

Gas-phase packed windows:

```
mmml umbrella-sample \
  --checkpoint model.json \
  --structure seed.xyz \
  --config umbrella.yaml
mmml umbrella-mbar --input umbrella_windows.json -o pmf/
```

Explicit-solvent mechanical embedding uses `engine: hybrid_jaxmd` with a PSF/PDB/box and an ML region (reactive complex) while solvent remains MM. Snapshots store unbiased hybrid energies so MBAR does not need to rebuild the Hamiltonian.

S3.2 CHARMM ADUMB

ADUMB is enabled via PyCHARMM pre-dynamics lingo (reaction-coordinate umbrella with `ADUMBRXNCOR` in the CHARMM build). Typical artifacts: `ADUMB-WUNI.DAT`, `UMBCOR`, `RXNCOR_TRACE.DAT`. NAME tokens are ≤ 4 characters. Example trees live under `examples/m/` and the `nh3_ch3cl` documentation.

Table S2: (PLACEHOLDER) Extended free-energy provenance (SI).

Protocol	CV	Artifact root	Git SHA
umbrella+MBAR	ξ	`artifacts/.../pmf/`	—
ADUMB+WHAM	ξ / multi	`artifacts/.../adumb/`	—

S4 Workflow Provenance

Each directory that feeds a main-text figure or table should contain:

1. frozen inputs (`request.json` / job YAML);
2. `metrics.json` (numbers that enter the paper);
3. plot PNG/PDF;
4. `provenance.json` — git SHA, checkpoint hash, host/partition;
5. optional `proof.json` when under validation-campaign gates.

Recommended manuscript aggregator: `workflows/manuscript_hybrid_liquids/` writing `artifacts/manuscript_hybrid_liquids/<fig_or_table_id>/`.

Table S3: Figure / table $\leftrightarrow$ workflow map (main text).

Panel	Content	Workflow
Fig. 5	NVE traces	`dcm3_nve_cutoff_sweep`
Fig. 6	Rigid scans	`des_dimer_pair_scans`
Fig. 7	$\rho(t)$	`pbc_liquid_density_dyn`
Fig. 8	Umbrella PMF	umbrella examples / Menshutkin
Fig. 9	ADUMB profile	ADUMB examples / `nh3_ch3cl`
Table 3	Spoof parity	`jaxmd_cgenff_spoof_smoke`
Table 5	$\langle\rho\rangle$	density / methane workflows

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